In [35]:
```python
import pandas as pd
```

Reading csv

In [89]:
```python
pulseData1 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData1.csv")
```

In [90]:
```python
print(pulseData1.to_string())
```

```
     Duration  Pulse  Maxpulse  Calories
0          60    110       130     409.1
1          60    117       145     479.0
2          60    103       135     340.0
3          45    109       175     282.4
4          45    117       148     406.0
5          60    102       127     300.0
6          60    110       136     374.0
7          45    104       134     253.3
8          30    109       133     195.1
9          60     98       124     269.0
10         60    103       147     329.3
11         60    100       120     250.7
12         60    106       128     345.3
13         60    104       132     379.3
14         60     98       123     275.0
15         60     98       120     215.2
16         60    100       120     300.0
17         45     90       112       NaN
```

Reading json

In [92]:
```
pulseData_json = pd.read_json(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData1.json")
```

In [93]:
```
print(pulseData_json.to_string())
```

```
     Duration  Pulse  Maxpulse  Calories
0          60    110       130     409.1
1          60    117       145     479.0
2          60    103       135     340.0
3          45    109       175     282.4
4          45    117       148     406.0
5          60    102       127     300.5
6          60    110       136     374.0
7          45    104       134     253.3
8          30    109       133     195.1
9          60     98       124     269.0
10         60    103       147     329.3
11         60    100       120     250.7
12         60    106       128     345.3
13         60    104       132     379.3
14         60     98       123     275.0
15         60     98       120     215.2
16         60    100       120     300.0
17         45     90       112       NaN
```

In [39]:
```
print(pulse_data1.head(10))
```

```
   Duration  Pulse  Maxpulse  Calories
0        60    110       130     409.1
1        60    117       145     479.0
2        60    103       135     340.0
3        45    109       175     282.4
4        45    117       148     406.0
5        60    102       127     300.0
6        60    110       136     374.0
7        45    104       134     253.3
8        30    109       133     195.1
9        60     98       124     269.0
```

In [40]:
```
print(pulse_data1.tail(10))
```

```
     Duration  Pulse  Maxpulse  Calories
159        30     80       120     240.9
160        30     85       120     250.4
161        45     90       130     260.4
162        45     95       130     270.0
163        45    100       140     280.9
164        60    105       140     290.8
165        60    110       145     300.0
166        60    115       145     310.2
167        75    120       150     320.4
168        75    125       150     330.4
```

Information about the data

In [41]:
```
print(pulse_data1.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 169 entries, 0 to 168
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Duration  169 non-null    int64
 1   Pulse     169 non-null    int64
 2   Maxpulse  169 non-null    int64
 3   Calories  164 non-null    float64
dtypes: float64(1), int64(3)
memory usage: 5.4 KB
None
```

Data Cleaning

In [197]:
```
pulse_data = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [198]:
```
print(pulse_data.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 32 entries, 0 to 31
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Duration  32 non-null     int64
 1   Date      31 non-null     object
 2   Pulse     32 non-null     int64
 3   Maxpulse  32 non-null     int64
 4   Calories  30 non-null     float64
dtypes: float64(1), int64(3), object(1)
memory usage: 1.4+ KB
None
```

print(pulse_data.to_string())

Removing empty cells

using dropna() function. Removes the rows with empty value or NaN

In [196]:
```
pulse_data2 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv", header=None, na
```

In [190]:
```
pulse_data2_cleaned = pulse_data2.dropna()
```

In [191]:

```python
print(pulse_data2_cleaned.to_string())
```

```
           0             1      2         3         4
0   Duration          Date  Pulse  Maxpulse  Calories
1         60  '2020/12/01'    110       130     409.1
2         60  '2020/12/02'    117       145     479.0
3         60  '2020/12/03'    103       135     340.0
4         45  '2020/12/04'    109       175     282.4
5         45  '2020/12/05'    117       148     406.0
6         60  '2020/12/06'    102       127     300.0
7         60  '2020/12/07'    110       136     374.0
8        450  '2020/12/08'    104       134     253.3
9         30  '2020/12/09'    109       133     195.1
10        60  '2020/12/10'     98       124     269.0
11        60  '2020/12/11'    103       147     329.3
12        60  '2020/12/12'    100       120     250.7
13        60  '2020/12/12'    100       120     250.7
14        60  '2020/12/13'    106       128     345.3
15        60  '2020/12/14'    104       132     379.3
16        60  '2020/12/15'     98       123     275.0
17        60  '2020/12/16'     98       120     215.2
18        60  '2020/12/17'    100       120     300.0
20        60  '2020/12/19'    103       123     323.0
21        45  '2020/12/20'     97       125     243.0
22        60  '2020/12/21'    108       131     364.2
24        60  '2020/12/23'    130       101     300.0
25        45  '2020/12/24'    105       132     246.0
26        60  '2020/12/25'    102       126     334.5
27        60      20201226    100       120     250.0
28        60  '2020/12/27'     92       118     241.0
30        60  '2020/12/29'    100       132     280.0
31        60  '2020/12/30'    102       129     380.3
32        60  '2020/12/31'     92       115     243.0
```

Replacing empty values with fillna() function

In [192]:
```
my_data1 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [193]:
```
my_data1.fillna(130, inplace = True)
```

In [195]:

```python
print(my_data1.to_string())
```

```
    Duration          Date  Pulse  Maxpulse  Calories
0         60  '2020/12/01'    110       130     409.1
1         60  '2020/12/02'    117       145     479.0
2         60  '2020/12/03'    103       135     340.0
3         45  '2020/12/04'    109       175     282.4
4         45  '2020/12/05'    117       148     406.0
5         60  '2020/12/06'    102       127     300.0
6         60  '2020/12/07'    110       136     374.0
7        450  '2020/12/08'    104       134     253.3
8         30  '2020/12/09'    109       133     195.1
9         60  '2020/12/10'     98       124     269.0
10        60  '2020/12/11'    103       147     329.3
11        60  '2020/12/12'    100       120     250.7
12        60  '2020/12/12'    100       120     250.7
13        60  '2020/12/13'    106       128     345.3
14        60  '2020/12/14'    104       132     379.3
15        60  '2020/12/15'     98       123     275.0
16        60  '2020/12/16'     98       120     215.2
17        60  '2020/12/17'    100       120     300.0
18        45  '2020/12/18'     90       112     130.0
19        60  '2020/12/19'    103       123     323.0
20        45  '2020/12/20'     97       125     243.0
21        60  '2020/12/21'    108       131     364.2
22        45           130    100       119     282.0
23        60  '2020/12/23'    130       101     300.0
24        45  '2020/12/24'    105       132     246.0
25        60  '2020/12/25'    102       126     334.5
26        60      20201226    100       120     250.0
27        60  '2020/12/27'     92       118     241.0
28        60  '2020/12/28'    103       132     130.0
29        60  '2020/12/29'    100       132     280.0
30        60  '2020/12/30'    102       129     380.3
31        60  '2020/12/31'     92       115     243.0
```

Replace Only for Specified Columns

In [201]:
```
my_data2 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [202]:
```
my_data2["Calories"].fillna(130, inplace=True)
```

In [203]:

```python
print(my_data2.to_string())
```

```
    Duration          Date  Pulse  Maxpulse  Calories
0         60  '2020/12/01'    110       130     409.1
1         60  '2020/12/02'    117       145     479.0
2         60  '2020/12/03'    103       135     340.0
3         45  '2020/12/04'    109       175     282.4
4         45  '2020/12/05'    117       148     406.0
5         60  '2020/12/06'    102       127     300.0
6         60  '2020/12/07'    110       136     374.0
7        450  '2020/12/08'    104       134     253.3
8         30  '2020/12/09'    109       133     195.1
9         60  '2020/12/10'     98       124     269.0
10        60  '2020/12/11'    103       147     329.3
11        60  '2020/12/12'    100       120     250.7
12        60  '2020/12/12'    100       120     250.7
13        60  '2020/12/13'    106       128     345.3
14        60  '2020/12/14'    104       132     379.3
15        60  '2020/12/15'     98       123     275.0
16        60  '2020/12/16'     98       120     215.2
17        60  '2020/12/17'    100       120     300.0
18        45  '2020/12/18'     90       112     130.0
19        60  '2020/12/19'    103       123     323.0
20        45  '2020/12/20'     97       125     243.0
21        60  '2020/12/21'    108       131     364.2
22        45           NaN    100       119     282.0
23        60  '2020/12/23'    130       101     300.0
24        45  '2020/12/24'    105       132     246.0
25        60  '2020/12/25'    102       126     334.5
26        60      20201226    100       120     250.0
27        60  '2020/12/27'     92       118     241.0
28        60  '2020/12/28'    103       132     130.0
29        60  '2020/12/29'    100       132     280.0
30        60  '2020/12/30'    102       129     380.3
31        60  '2020/12/31'     92       115     243.0
```

Replacing with mean, median and mode

In [204]:
```
my_data3 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [205]:
```
x = my_data3["Calories"].mean()
```

In [206]:
```
my_data3["Calories"].fillna(x, inplace = True)
```

In [207]:

```python
print(my_data3.to_string())
```

```
    Duration          Date  Pulse  Maxpulse  Calories
0         60  '2020/12/01'    110       130    409.10
1         60  '2020/12/02'    117       145    479.00
2         60  '2020/12/03'    103       135    340.00
3         45  '2020/12/04'    109       175    282.40
4         45  '2020/12/05'    117       148    406.00
5         60  '2020/12/06'    102       127    300.00
6         60  '2020/12/07'    110       136    374.00
7        450  '2020/12/08'    104       134    253.30
8         30  '2020/12/09'    109       133    195.10
9         60  '2020/12/10'     98       124    269.00
10        60  '2020/12/11'    103       147    329.30
11        60  '2020/12/12'    100       120    250.70
12        60  '2020/12/12'    100       120    250.70
13        60  '2020/12/13'    106       128    345.30
14        60  '2020/12/14'    104       132    379.30
15        60  '2020/12/15'     98       123    275.00
16        60  '2020/12/16'     98       120    215.20
17        60  '2020/12/17'    100       120    300.00
18        45  '2020/12/18'     90       112    304.68
19        60  '2020/12/19'    103       123    323.00
20        45  '2020/12/20'     97       125    243.00
21        60  '2020/12/21'    108       131    364.20
22        45           NaN    100       119    282.00
23        60  '2020/12/23'    130       101    300.00
24        45  '2020/12/24'    105       132    246.00
25        60  '2020/12/25'    102       126    334.50
26        60      20201226    100       120    250.00
27        60  '2020/12/27'     92       118    241.00
28        60  '2020/12/28'    103       132    304.68
29        60  '2020/12/29'    100       132    280.00
30        60  '2020/12/30'    102       129    380.30
31        60  '2020/12/31'     92       115    243.00
```

Cleaning Data of Wrong Format

In [208]:
```
my_data4 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [213]:
```
my_data4['Date'] = pd.to_datetime(my_data4['Date'], format='mixed') # format= mixed is passed because of NaN
```

In [212]:

```
print(my_data4.to_string())
```

```
    Duration       Date  Pulse  Maxpulse  Calories
0         60 2020-12-01    110       130     409.1
1         60 2020-12-02    117       145     479.0
2         60 2020-12-03    103       135     340.0
3         45 2020-12-04    109       175     282.4
4         45 2020-12-05    117       148     406.0
5         60 2020-12-06    102       127     300.0
6         60 2020-12-07    110       136     374.0
7        450 2020-12-08    104       134     253.3
8         30 2020-12-09    109       133     195.1
9         60 2020-12-10     98       124     269.0
10        60 2020-12-11    103       147     329.3
11        60 2020-12-12    100       120     250.7
12        60 2020-12-12    100       120     250.7
13        60 2020-12-13    106       128     345.3
14        60 2020-12-14    104       132     379.3
15        60 2020-12-15     98       123     275.0
16        60 2020-12-16     98       120     215.2
17        60 2020-12-17    100       120     300.0
18        45 2020-12-18     90       112       NaN
19        60 2020-12-19    103       123     323.0
20        45 2020-12-20     97       125     243.0
21        60 2020-12-21    108       131     364.2
22        45        NaT    100       119     282.0
23        60 2020-12-23    130       101     300.0
24        45 2020-12-24    105       132     246.0
25        60 2020-12-25    102       126     334.5
26        60 2020-12-26    100       120     250.0
27        60 2020-12-27     92       118     241.0
28        60 2020-12-28    103       132       NaN
29        60 2020-12-29    100       132     280.0
30        60 2020-12-30    102       129     380.3
31        60 2020-12-31     92       115     243.0
```

In [214]:

```
my_data4.dropna(subset=['Date'], inplace = True)
```

In [215]:
```python
print(my_data4.to_string())
```

```
    Duration       Date  Pulse  Maxpulse  Calories
0         60 2020-12-01    110       130     409.1
1         60 2020-12-02    117       145     479.0
2         60 2020-12-03    103       135     340.0
3         45 2020-12-04    109       175     282.4
4         45 2020-12-05    117       148     406.0
5         60 2020-12-06    102       127     300.0
6         60 2020-12-07    110       136     374.0
7        450 2020-12-08    104       134     253.3
8         30 2020-12-09    109       133     195.1
9         60 2020-12-10     98       124     269.0
10        60 2020-12-11    103       147     329.3
11        60 2020-12-12    100       120     250.7
12        60 2020-12-12    100       120     250.7
13        60 2020-12-13    106       128     345.3
14        60 2020-12-14    104       132     379.3
15        60 2020-12-15     98       123     275.0
16        60 2020-12-16     98       120     215.2
17        60 2020-12-17    100       120     300.0
18        45 2020-12-18     90       112       NaN
19        60 2020-12-19    103       123     323.0
20        45 2020-12-20     97       125     243.0
21        60 2020-12-21    108       131     364.2
23        60 2020-12-23    130       101     300.0
24        45 2020-12-24    105       132     246.0
25        60 2020-12-25    102       126     334.5
26        60 2020-12-26    100       120     250.0
27        60 2020-12-27     92       118     241.0
28        60 2020-12-28    103       132       NaN
29        60 2020-12-29    100       132     280.0
30        60 2020-12-30    102       129     380.3
31        60 2020-12-31     92       115     243.0
```

Fixing Wrong data(eg. typo)

eg. In the data above, the typo in row 7: 450 can be corrected by replacing by inserting 45.

In [216]:

```
my_data5 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [217]:

```
my_data5.loc[7, 'Duration'] = 45
```

In [218]:

```
print(my_data5.to_string())
```

```
    Duration          Date  Pulse  Maxpulse  Calories
0         60  '2020/12/01'    110       130     409.1
1         60  '2020/12/02'    117       145     479.0
2         60  '2020/12/03'    103       135     340.0
3         45  '2020/12/04'    109       175     282.4
4         45  '2020/12/05'    117       148     406.0
5         60  '2020/12/06'    102       127     300.0
6         60  '2020/12/07'    110       136     374.0
7         45  '2020/12/08'    104       134     253.3
8         30  '2020/12/09'    109       133     195.1
9         60  '2020/12/10'     98       124     269.0
10        60  '2020/12/11'    103       147     329.3
11        60  '2020/12/12'    100       120     250.7
12        60  '2020/12/12'    100       120     250.7
13        60  '2020/12/13'    106       128     345.3
14        60  '2020/12/14'    104       132     379.3
15        60  '2020/12/15'     98       123     275.0
16        60  '2020/12/16'     98       120     215.2
17        60  '2020/12/17'    100       120     300.0
18        45  '2020/12/18'     90       112       NaN
19        60  '2020/12/19'    103       123     323.0
20        45  '2020/12/20'     97       125     243.0
21        60  '2020/12/21'    108       131     364.2
22        45           NaN    100       119     282.0
23        60  '2020/12/23'    130       101     300.0
24        45  '2020/12/24'    105       132     246.0
25        60  '2020/12/25'    102       126     334.5
26        60      20201226    100       120     250.0
27        60  '2020/12/27'     92       118     241.0
28        60  '2020/12/28'    103       132       NaN
29        60  '2020/12/29'    100       132     280.0
30        60  '2020/12/30'    102       129     380.3
31        60  '2020/12/31'     92       115     243.0
```

For small data sets, it easier to spot a error and replace it, but for big data sets, its not. To replace wrong data for larger data sets, we can as follows:

In [219]:

```
my_data6 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [221]:

```
for i in my_data6.index:
    if my_data6.loc[i, "Duration"] > 120:
        my_data6.loc[i, "Duration"] = 120
```

In [222]:

```python
print(my_data6.to_string())
```

```
    Duration          Date  Pulse  Maxpulse  Calories
0         60  '2020/12/01'    110       130     409.1
1         60  '2020/12/02'    117       145     479.0
2         60  '2020/12/03'    103       135     340.0
3         45  '2020/12/04'    109       175     282.4
4         45  '2020/12/05'    117       148     406.0
5         60  '2020/12/06'    102       127     300.0
6         60  '2020/12/07'    110       136     374.0
7        120  '2020/12/08'    104       134     253.3
8         30  '2020/12/09'    109       133     195.1
9         60  '2020/12/10'     98       124     269.0
10        60  '2020/12/11'    103       147     329.3
11        60  '2020/12/12'    100       120     250.7
12        60  '2020/12/12'    100       120     250.7
13        60  '2020/12/13'    106       128     345.3
14        60  '2020/12/14'    104       132     379.3
15        60  '2020/12/15'     98       123     275.0
16        60  '2020/12/16'     98       120     215.2
17        60  '2020/12/17'    100       120     300.0
18        45  '2020/12/18'     90       112       NaN
19        60  '2020/12/19'    103       123     323.0
20        45  '2020/12/20'     97       125     243.0
21        60  '2020/12/21'    108       131     364.2
22        45           NaN    100       119     282.0
23        60  '2020/12/23'    130       101     300.0
24        45  '2020/12/24'    105       132     246.0
25        60  '2020/12/25'    102       126     334.5
26        60      20201226    100       120     250.0
27        60  '2020/12/27'     92       118     241.0
28        60  '2020/12/28'    103       132       NaN
29        60  '2020/12/29'    100       132     280.0
30        60  '2020/12/30'    102       129     380.3
31        60  '2020/12/31'     92       115     243.0
```

or by removing rows

In [223]:
```
my_data7 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [225]:
```
for x in my_data7.index:
    if my_data7.loc[x, "Duration"] > 120:
        my_data7.drop(x, inplace = True)
```

In [226]:

```python
print(my_data7.to_string())
```

```
    Duration          Date  Pulse  Maxpulse  Calories
0         60  '2020/12/01'    110       130     409.1
1         60  '2020/12/02'    117       145     479.0
2         60  '2020/12/03'    103       135     340.0
3         45  '2020/12/04'    109       175     282.4
4         45  '2020/12/05'    117       148     406.0
5         60  '2020/12/06'    102       127     300.0
6         60  '2020/12/07'    110       136     374.0
8         30  '2020/12/09'    109       133     195.1
9         60  '2020/12/10'     98       124     269.0
10        60  '2020/12/11'    103       147     329.3
11        60  '2020/12/12'    100       120     250.7
12        60  '2020/12/12'    100       120     250.7
13        60  '2020/12/13'    106       128     345.3
14        60  '2020/12/14'    104       132     379.3
15        60  '2020/12/15'     98       123     275.0
16        60  '2020/12/16'     98       120     215.2
17        60  '2020/12/17'    100       120     300.0
18        45  '2020/12/18'     90       112       NaN
19        60  '2020/12/19'    103       123     323.0
20        45  '2020/12/20'     97       125     243.0
21        60  '2020/12/21'    108       131     364.2
22        45           NaN    100       119     282.0
23        60  '2020/12/23'    130       101     300.0
24        45  '2020/12/24'    105       132     246.0
25        60  '2020/12/25'    102       126     334.5
26        60      20201226    100       120     250.0
27        60  '2020/12/27'     92       118     241.0
28        60  '2020/12/28'    103       132       NaN
29        60  '2020/12/29'    100       132     280.0
30        60  '2020/12/30'    102       129     380.3
31        60  '2020/12/31'     92       115     243.0
```

Removing Duplicates

In [227]:

```
my_data8 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData2.csv")
```

In [228]:
```
print(my_data8.duplicated())
```

```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12     True
13    False
14    False
15    False
16    False
17    False
18    False
19    False
20    False
21    False
22    False
23    False
24    False
25    False
26    False
27    False
28    False
29    False
30    False
31    False
dtype: bool
```

In [230]:
```
my_data8.drop_duplicates(inplace = True)
```

In [231]:
```python
print(my_data8.to_string())
```

```
    Duration          Date  Pulse  Maxpulse  Calories
0         60  '2020/12/01'    110       130     409.1
1         60  '2020/12/02'    117       145     479.0
2         60  '2020/12/03'    103       135     340.0
3         45  '2020/12/04'    109       175     282.4
4         45  '2020/12/05'    117       148     406.0
5         60  '2020/12/06'    102       127     300.0
6         60  '2020/12/07'    110       136     374.0
7        450  '2020/12/08'    104       134     253.3
8         30  '2020/12/09'    109       133     195.1
9         60  '2020/12/10'     98       124     269.0
10        60  '2020/12/11'    103       147     329.3
11        60  '2020/12/12'    100       120     250.7
13        60  '2020/12/13'    106       128     345.3
14        60  '2020/12/14'    104       132     379.3
15        60  '2020/12/15'     98       123     275.0
16        60  '2020/12/16'     98       120     215.2
17        60  '2020/12/17'    100       120     300.0
18        45  '2020/12/18'     90       112       NaN
19        60  '2020/12/19'    103       123     323.0
20        45  '2020/12/20'     97       125     243.0
21        60  '2020/12/21'    108       131     364.2
22        45           NaN    100       119     282.0
23        60  '2020/12/23'    130       101     300.0
24        45  '2020/12/24'    105       132     246.0
25        60  '2020/12/25'    102       126     334.5
26        60      20201226    100       120     250.0
27        60  '2020/12/27'     92       118     241.0
28        60  '2020/12/28'    103       132       NaN
29        60  '2020/12/29'    100       132     280.0
30        60  '2020/12/30'    102       129     380.3
31        60  '2020/12/31'     92       115     243.0
```

Data Correlations

Finding relationships

In [232]:
```
df1 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData1.csv")
```

In [233]:
```
df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 169 entries, 0 to 168
Data columns (total 4 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Duration  169 non-null    int64
 1   Pulse     169 non-null    int64
 2   Maxpulse  169 non-null    int64
 3   Calories  164 non-null    float64
dtypes: float64(1), int64(3)
memory usage: 5.4 KB
```

In [235]:
```
print(df1.corr())
```

```
          Duration     Pulse  Maxpulse  Calories
Duration  1.000000 -0.155408  0.009403  0.922717
Pulse    -0.155408  1.000000  0.786535  0.025121
Maxpulse  0.009403  0.786535  1.000000  0.203813
Calories  0.922717  0.025121  0.203813  1.000000
```

Plotting

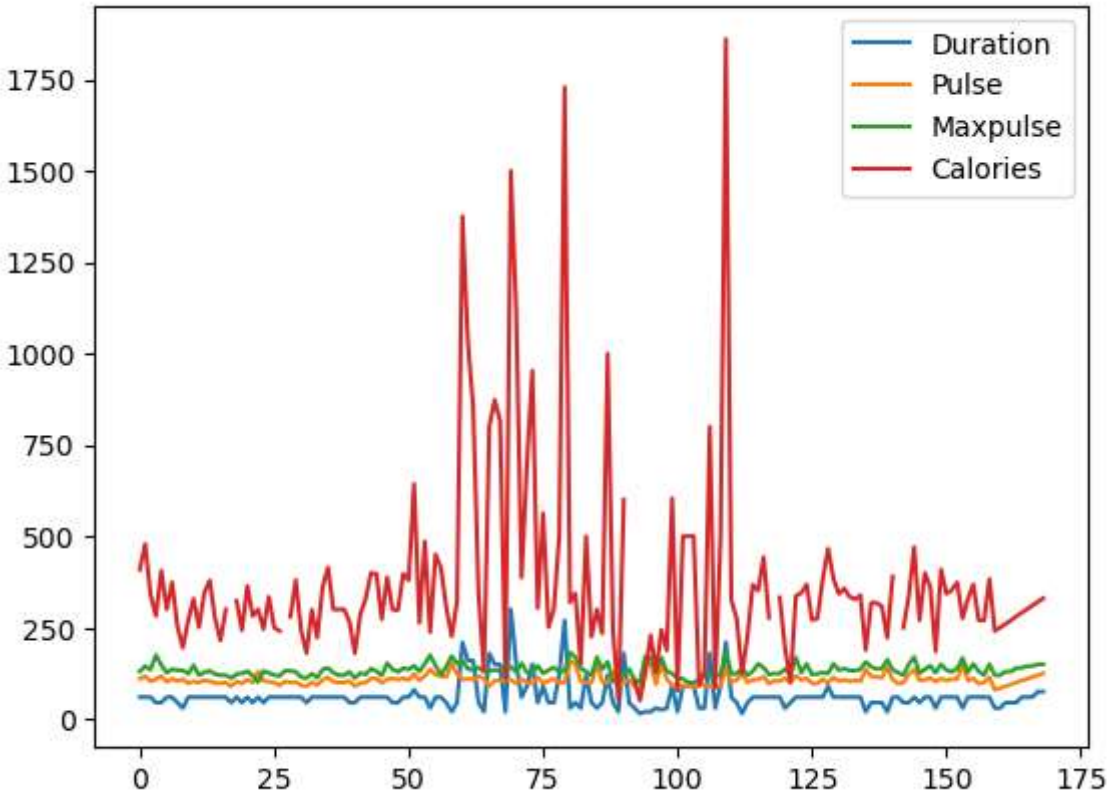
With pandas, we can use pyplot submodule of the Matplotlib library to visualize

In [ ]:
```
conda install matplotlib
```

In [1]:

```python
import pandas as pd
import matplotlib.pyplot as plt
df1 = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData1.csv")

df1.plot()
plt.show()

```

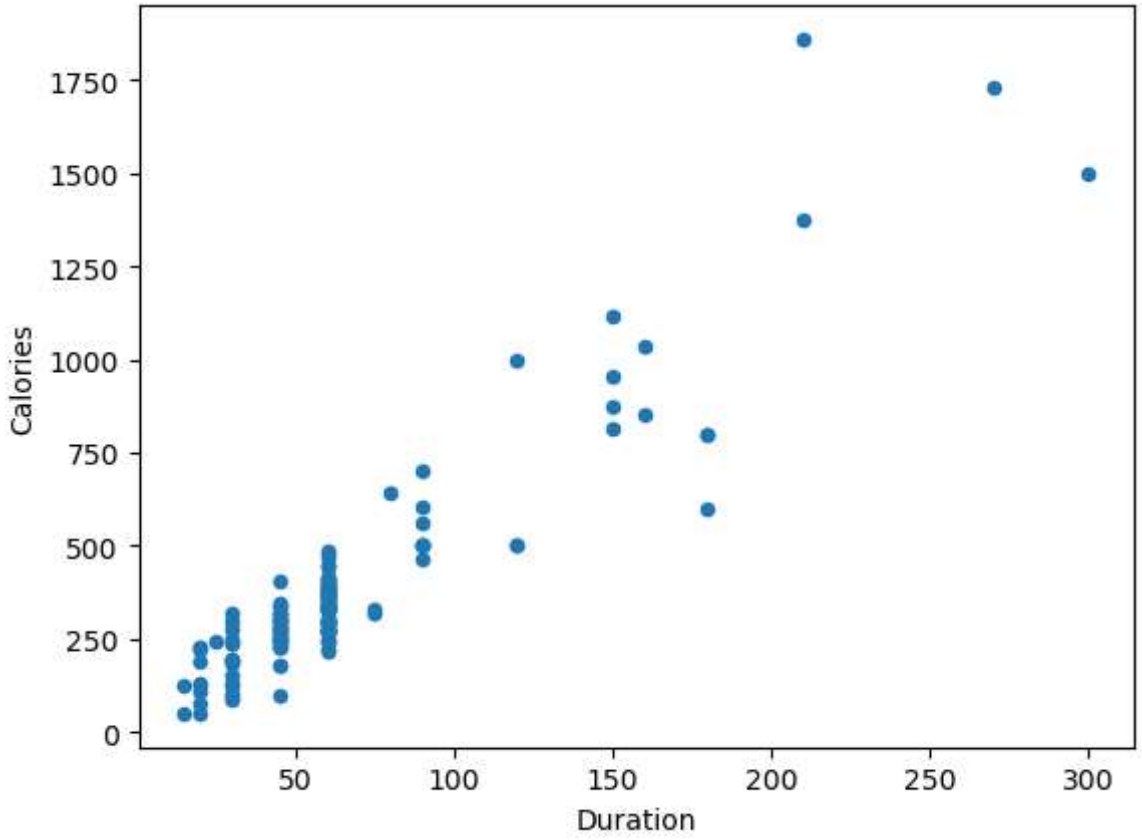
```
C:\Users\35844\anaconda3\Lib\site-packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version '1.
3.6' or newer of 'bottleneck' (version '1.3.5' currently installed).
  from pandas.core import (
```



In [1]:

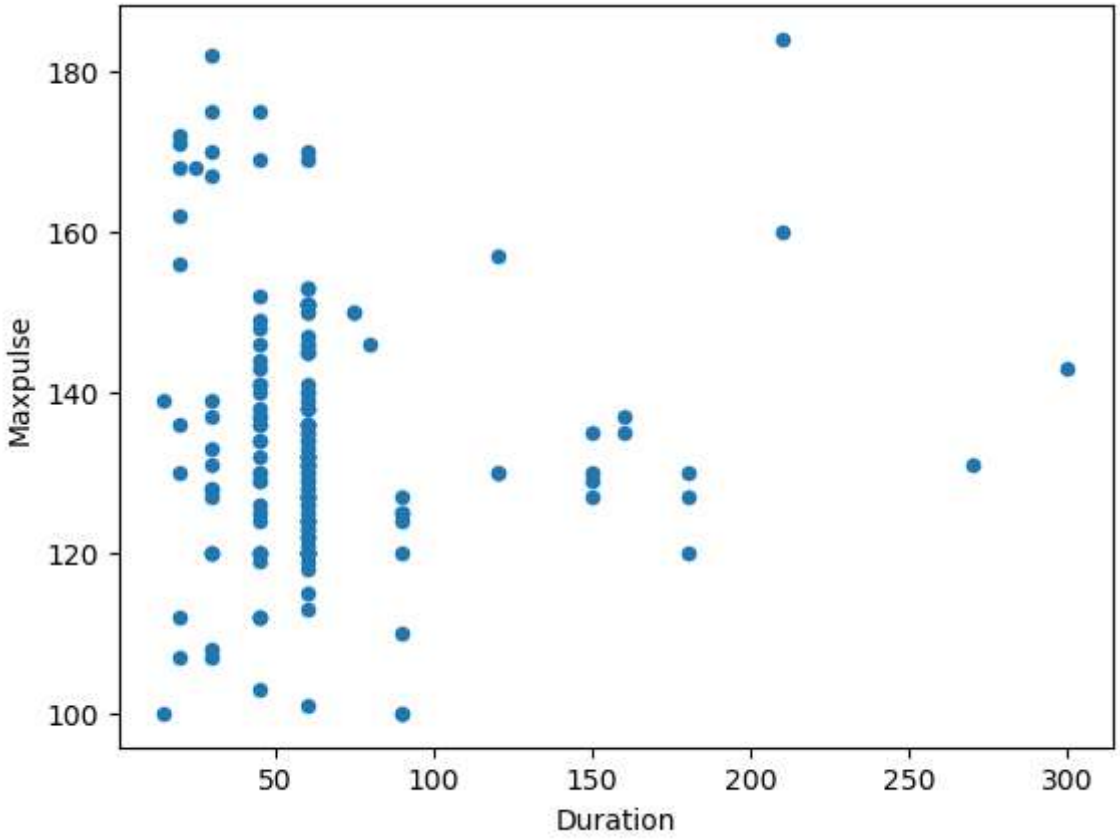
```python
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData1.csv")

df.plot(kind = 'scatter', x = 'Duration', y = 'Calories')

plt.show()
```

```
C:\Users\35844\anaconda3\Lib\site-packages\pandas\core\arrays\masked.py:60: UserWarning: Pandas requires version '1.
3.6' or newer of 'bottleneck' (version '1.3.5' currently installed).
  from pandas.core import (
```



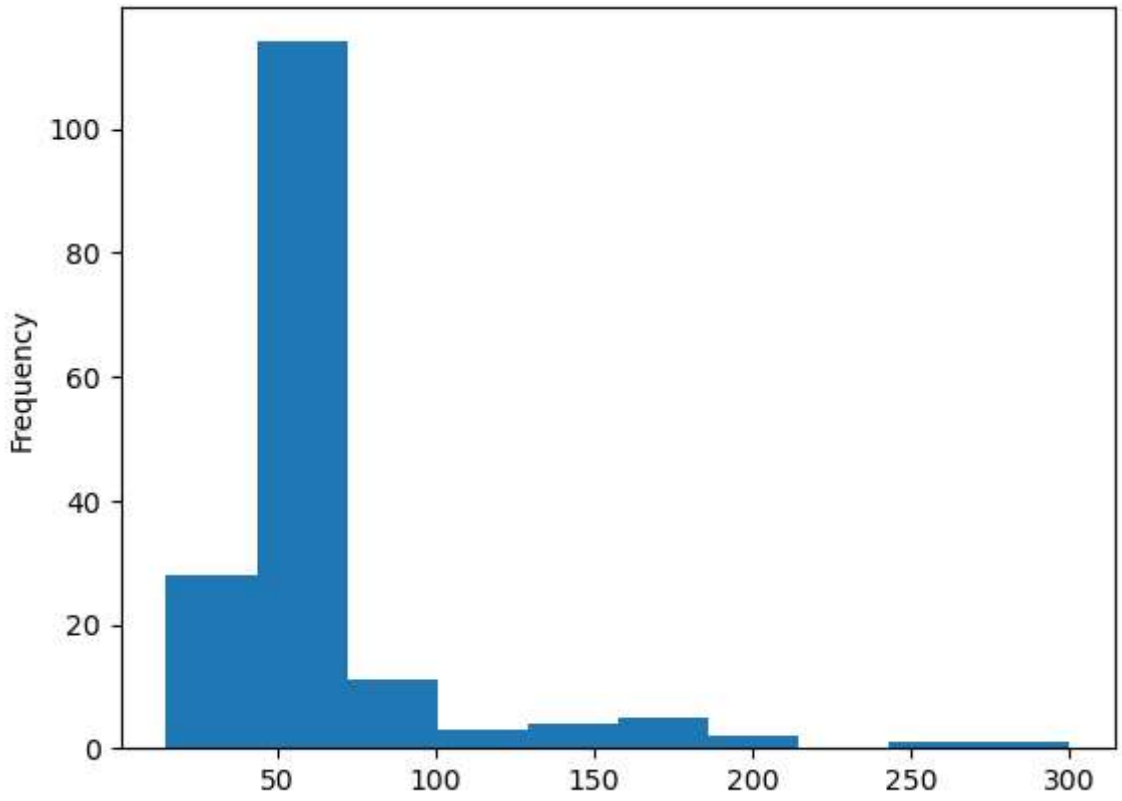
In [2]:

```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\35844\OneDrive\Desktop\Git_Page\mypage\data\pulseData1.csv")

df.plot(kind = 'scatter', x = 'Duration', y = 'Maxpulse')
plt.show()
```



In [3]:

```
df["Duration"].plot(kind = 'hist')
plt.show()
```



In [ ]:

```

```